BATTER-MoE: A Sparse Mixture-of-Experts Model for Accurate and Efficient Remaining Useful Life Prediction of Lithium-Ion Batteries

Xuan Yi, Jianmao Xiao*, Gang Lei, Xin Hu, Musheng Wu, Bo Xu, and Chuying Ouyang

***Abstract*—Accurately estimating the remaining useful life (RUL) of batteries from degradation trajectories is critical for system safety but remains challenging due to the complex and heterogeneous nature of degradation processes. Existing dense predictors achieve strong accuracy but process heterogeneous degradation trajectories through a shared full-capacity pathway, which may limit deployment flexibility in resource-constrained settings due to their computational cost. To address this limitation, we propose BATTER-MoE, a sparse Mixture-of-Experts (MoE) framework that couples battery-oriented temporal representation learning with sparse conditional computation. The framework combines time-aware multi-scale tokenization and cross-scale feature fusion with temporal reweighting, a sparse MoE backbone with RoPE-driven self-attention and a shared expert mechanism, and time-aware pooling for prediction. Experimental results on three public datasets (NASA, GOTION, and TJU), covering different chemistries and capacities, demonstrate that BATTER-MoE achieves competitive performance compared with representative baselines. Across multiple settings, it achieves the best or tied-best MAE/RMSE in most evaluations while maintaining a competitive balance between predictive accuracy and computational cost. This study suggests that sparse conditional computation is a promising direction for battery prognostics when both predictive accuracy and computational efficiency are important.**

***Index Terms*—Remaining useful life (RUL), mixture of experts (MoE), lithium-ion batteries, Transformers.**

I. Introduction

Driven by global carbon-neutrality goals, electric vehicles and energy storage systems are expanding rapidly. Lithium-ion batteries are a core component of these systems, and accurate health management is essential for ensuring system safety and reducing operational costs [1]. In this context, precise prediction of the remaining useful life (RUL) of batteries is a key function of battery management systems (BMS), because it supports maintenance planning, reliability assessment, and safe operation throughout the service life of the system [2].

Nevertheless, battery RUL prediction remains challenging because battery aging is governed by coupled nonlinear electrochemical mechanisms and is influenced by chemistry, operating conditions, and usage history [3]. Capacity regeneration during rest periods can introduce local fluctuations into measured degradation trajectories [4]. These non-monotonic variations may obscure the long-term aging trend and complicate direct sequence modeling for RUL prediction [5]. Battery aging data are also heterogeneous across cell chemistries, rated capacities, and sensing modalities [6]. Recent studies on full-life-cycle battery health assessment have further highlighted the need for prognostic models that generalize across diverse data sources and observation settings [7]. Battery RUL prediction is therefore not only a temporal forecasting task but also a heterogeneous degradation modeling problem in which decline rates vary across aging stages, and turning points often reflect changes in degradation behavior.

Recent data-driven methods, especially deep learning models, have substantially improved predictive accuracy in battery prognostics [8]. Transformers have shown strong capability for long-range sequence modeling [9]. Modern state-space models, such as Mamba, have also shown promise for battery RUL modeling [10]. However, most existing deep RUL predictors still follow a shared dense-computation paradigm in which every input is processed by the same full-capacity backbone, regardless of whether smooth long-term fade, regeneration-like fluctuations, or dataset-specific degradation characteristics dominate the observed trajectory. For battery prognostics, this assumption fundamentally mismatches the heterogeneous and evolving nature of battery degradation, where different chemistries, capacities, and operating conditions give rise to distinct degradation patterns, and even a single trajectory may exhibit regime shifts, such as accelerated decline or regeneration effects.

Accordingly, the central issue is not simply whether a larger model can achieve lower error, but whether model capacity should be dynamically allocated to the input's degradation regime. Motivated by this consideration, we investigate sparse Mixture-of-Experts (MoE) modeling for battery RUL prediction. Sparse MoE decouples total model capacity from the amount of computation activated for each input [11]. For battery sequences, this property is particularly important because it enables input-dependent capacity allocation, allowing different experts to specialize in distinct degradation regimes such as smooth fading, accelerated decline, or regeneration-affected fluctuations. In this view, the value of sparse conditional computation extends beyond computational efficiency;

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more fundamentally, it provides a mechanism for aligning expert specialization with heterogeneous degradation patterns.

Motivated by this, we propose **BATTER-MoE** (Battery-Aware Time-series Transformer with Expert Routing), a sparse framework for battery RUL prediction, rather than treating MoE as a standalone backbone replacement. BATTER-MoE couples sparse expert routing with battery-oriented temporal representation learning. Specifically, the model performs *time-aware multi-scale tokenization* to encode degradation information at different temporal resolutions, applies *Cross-Scale Squeeze-Excitation Attention (CS-SE attention)* for feature fusion [12], and uses a *Cross-Time Reweighting (CT-Reweighting)* module to enhance informative temporal patterns before backbone encoding. Within the encoder, RoPE-driven self-attention encodes temporal position information, while the sparse MoE block combines routed experts via a shared expert mechanism. A *time-aware pooling* strategy then focuses prediction on the latest valid timestamp. Compared with a recent MoE-based battery RUL study that combines MLP experts with temporal feature composition, BATTER-MoE embeds sparse routing in a Transformer-style backbone and further incorporates a battery-oriented temporal front end together with a shared expert mechanism [13].

We conduct comprehensive experiments across three public datasets — NASA, GOTION, and TJU — covering different chemistries, cell capacities, and input modalities. In this work, RUL estimation is based on one-step-ahead capacity prediction under different observation settings. Experimental results show that BATTER-MoE achieves competitive predictive accuracy across heterogeneous settings while providing a better balance between predictive accuracy and computational cost than representative dense baselines. These findings indicate that sparse conditional computation is a promising approach for battery RUL prediction when both degradation heterogeneity and computational efficiency must be considered.

The main contributions of this work are summarized as follows:

- To address the heterogeneity of battery degradation trajectories, we propose BATTER-MoE. This sparse battery prognostics framework introduces input-dependent capacity allocation through conditional expert routing for more adaptive modeling of heterogeneous degradation.
- To better capture degradation information across temporal scales and emphasize informative aging patterns, we develop a unified temporal modeling framework that integrates time-aware multi-scale tokenization, Cross-Scale Squeeze-Excitation attention, Cross-Time Reweighting, and time-aware readout.
- Extensive experiments on three heterogeneous public datasets with different chemistries, capacities, and input modalities show that the proposed framework achieves competitive predictive performance while maintaining a favorable balance between predictive accuracy and computational cost.

II. Related Work

In recent years, data-driven methods have become a major research direction in lithium-ion battery RUL prediction. In this area, machine learning (ML) [14] and deep learning (DL) [15] have demonstrated strong capability in modeling complex, nonlinear degradation processes. Before the widespread adoption of deep learning, traditional ML methods were extensively explored. Anton et al. [16] employed support vector regression (SVR) for state estimation; Li et al. [17] combined minimum entropy with the relevance vector machine (RVM) for health monitoring; and Liu et al. [18] adopted Gaussian process regression (GPR) to quantify predictive uncertainty. However, such methods typically rely on handcrafted features and simplified statistical assumptions, which limit their ability to capture long-range temporal dependencies and complex degradation behaviors such as capacity regeneration.

With the development of deep representation learning, recurrent neural networks (RNNs) and convolutional neural networks (CNNs) have emerged as two main lines of research. Zhang et al. [19] employed LSTM networks to alleviate long-range dependency issues. In contrast, Yang et al. [20] constructed images from single-cycle data and used CNNs for ultra-early lifetime prediction. Hybrid architectures that combine multiple models have also been investigated [21]. However, existing RNN- and CNN-based methods still face challenges in modeling long-range dependencies under complex degradation patterns.

To further enhance long-sequence modeling, Transformers and recent state space models (SSMs), such as Mamba, have attracted increasing attention. Several works have customized Transformer architectures for RUL prediction [22] and proposed patch-based variants [23]. For SSMs, Huang et al. [10] and Wang et al. [24] have demonstrated their potential for extracting complex temporal features. Recent battery prognostics studies have also examined non-monotonic health trajectories by separating long-term degradation from fluctuation-related dynamics, suggesting that regeneration-related variations may contain structured information rather than mere measurement noise [5]. More broadly, related studies in prognostics and battery-system analysis have underscored the importance of characterizing multi-scale degradation behavior, interacting evolution patterns, and uncertainty in complex systems [25]–[27]. Taken together, these studies suggest that effective prognostic models should capture not only global degradation trends but also local fluctuations, evolving interactions, and heterogeneous operating conditions.

Despite these advances, many recent battery RUL models still rely on dense, single-backbone architectures, in which all inputs are processed through the same computational pathway. While such designs can achieve strong predictive performance, they do not explicitly exploit input-dependent conditional computation to adapt model capacity to heterogeneous degradation regimes. This limitation is particularly relevant in battery prognostics, where degradation trajectories may differ substantially across chemistries, capacities, and operating conditions.

Mixture-of-Experts (MoE) provides a practical mechanism for sparse conditional computation by routing different inputs to different expert subnetworks. In battery RUL prediction, recent studies have begun to explore expert-based architectures. For example, Chen et al. [28] combined attention with expert routing for lithium-ion battery RUL prediction. Yi et

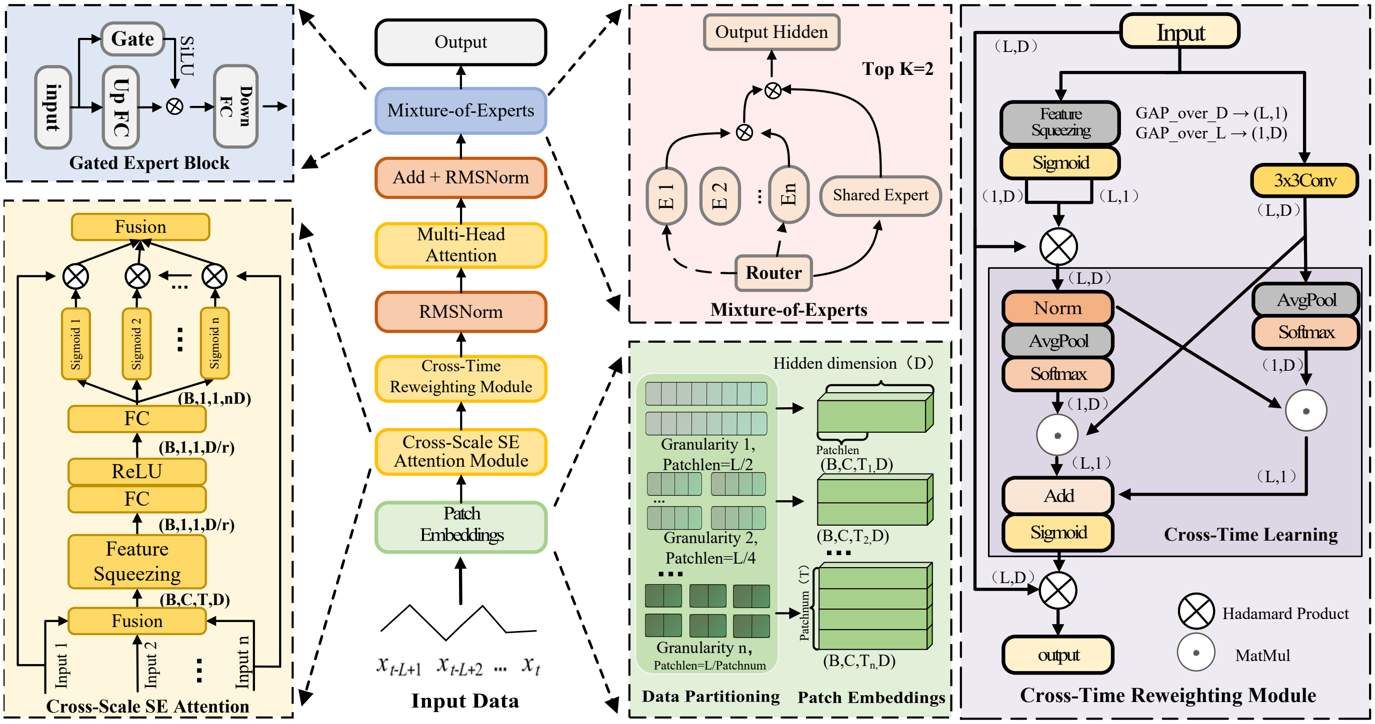

Fig. 1. Overall framework of BATTER-MoE, illustrating the cross-scale SE attention module, cross-time reweighting module, gated expert block, and the Mixture of Experts routing mechanism.

al. [13] proposed an MoE-based framework with multilayer perceptron expert networks and temporal feature composition for battery RUL prediction. These studies indicate that expert-based conditional modeling warrants further investigation in battery prognostics. However, in existing battery RUL studies, MoE has typically been incorporated as one component within a broader prediction pipeline. By comparison, less attention has been paid to jointly combining sparse routing with explicit physical-time modeling, cross-scale feature interaction, and time-aware readout within a unified Transformer-style framework. Motivated by this gap, we propose **BATTER-MoE**, which integrates sparse routing with battery-oriented temporal representation learning within a unified framework for heterogeneous battery degradation modeling and RUL prediction.

III. Methodology

A. Overall Architecture of BATTER-MoE

As shown in Fig. 1, **BATTER-MoE** is designed for battery aging prediction. Before entering the backbone, the model applies *time-aware multi-scale tokenization* to construct a token sequence with associated temporal positions. Within the backbone, RoPE-driven self-attention and sparse MoE experts jointly model multi-scale temporal patterns. Finally, *time-aware pooling* focuses on the latest valid timestamp. The overall pipeline is summarized in (1):

$$\mathbf{X} \xrightarrow{\mathcal{T}(\cdot;\mathcal{P})} (\mathbf{Z}, \mathrm{pos}, \mathrm{mask}) \xrightarrow{\mathcal{B}} \mathbf{H} \xrightarrow{\mathcal{P}_t} \mathbf{h} \xrightarrow{\mathbf{W}} \hat{y}. \tag{1}$$

where $\mathcal{T}(\cdot;\mathcal{P})$ denotes the time-aware multi-scale tokenization module parameterized by $\mathcal{P}$, which converts the raw sequence $\mathbf{X}$ into token embeddings $\mathbf{Z}$ with associated temporal-position encodings pos and a validity mask mask; $\mathcal{B}$ is the backbone encoder, which consists of RoPE-based self-attention and sparse-plus-shared MoE layers and produces contextual representations $\mathbf{H}$; $\mathcal{P}_t$ is the time-aware pooling operator that aggregates $\mathbf{H}$ around the latest valid timestamp into a compact feature vector $\mathbf{h}$; and $\mathbf{W}$ is a linear prediction head that maps $\mathbf{h}$ to the next-step prediction target $\hat{y}$.

B. Time-aware Multi-Scale Tokenization

We consider an input sequence $\mathbf{X} \in \mathbb{R}^{L\times C}$, where L denotes the sequence length and C denotes the number of input channels. Let $\mathcal{P} = \{p_1, \ldots, p_K\}$ be a set of patch lengths for non-overlapping patching, where each scale p_i uses a stride equal to its patch length. Each patch is then projected into an H-dimensional hidden space.

1) Multi-Scale Non-Overlapping Patching: For scale i and patch index $n = 0, \ldots, N_i - 1$, the n-th segment is defined as $\mathbf{S}_n^{(i)} = \mathbf{X}[np_i : (n+1)p_i, :] \in \mathbb{R}^{p_i \times C}$. Its token is then obtained by a linear projection:

$$\mathbf{t}_n^{(i)} = \mathrm{vec}\big(\mathbf{S}_n^{(i)}\big)\mathbf{W}_i + \mathbf{b}_i \in \mathbb{R}^H, \qquad \mathbf{W}_i \in \mathbb{R}^{(p_i C)\times H}. \tag{2}$$

If scale embeddings are enabled, we update $\mathbf{t}_n^{(i)} \leftarrow \mathbf{t}_n^{(i)} + \mathbf{e}_i$ with $\mathbf{e}_i \in \mathbb{R}^H$. We denote $\mathbf{T}_i = [\mathbf{t}_0^{(i)}; \ldots; \mathbf{t}_{N_i-1}^{(i)}] \in \mathbb{R}^{N_i\times H}$, where $N_i = \lfloor L/p_i \rfloor$. The center timestamp is $\mathrm{pos}_n^{(i)} = np_i + \lfloor (p_i - 1)/2 \rfloor$ for $n = 0, \ldots, N_i - 1$, and the validity mask $\mathrm{mask}_n^{(i)}$ is obtained as the minimum of the corresponding input-mask entries (or set to 1 if no mask is provided).

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2) Cross-Scale Squeeze-Excitation Attention (CS-SE attention): We first construct a context vector for each scale and feed it into a shared MLP to produce per-branch channel weights, which are then written back to the tokens channel-wise:

$$\begin{aligned} \mathbf{c}_i &= \tfrac{1}{N_i}\sum_{n=0}^{N_i-1}\mathbf{t}_n^{(i)}, \qquad \bar{\mathbf{c}} = \sum_{i=1}^{K}\mathbf{c}_i \in \mathbb{R}^{1\times H},\\ \tilde{\mathbf{a}} &= \sigma\Big(\mathrm{ReLU}\big(\bar{\mathbf{c}}\,\mathbf{W}_1\big)\,\mathbf{W}_2\Big) \in \mathbb{R}^{1\times (KH)},\\ \mathbf{A} &= \mathrm{reshape}_{K\times H}\big(\tilde{\mathbf{a}}\big) \in \mathbb{R}^{K\times H}, \end{aligned} \tag{3}$$

where $\mathbf{W}_1 \in \mathbb{R}^{H\times(H/r)}$ and $\mathbf{W}_2 \in \mathbb{R}^{(H/r)\times(KH)}$, r is the reduction ratio, and σ denotes the sigmoid function. Let the i-th row be $\mathbf{a}_i = \mathbf{A}_{i,:} \in \mathbb{R}^{1\times H}$. For the token sequence at scale i, $\mathbf{T}_i \in \mathbb{R}^{N_i\times H}$, the channel-wise writeback is

$$\hat{\mathbf{T}}_i \;=\; \mathbf{T}_i \odot \big(\mathbf{1}_{N_i}\mathbf{a}_i\big) \;=\; \big[\,\mathbf{t}_n^{(i)} \odot \mathbf{a}_i\,\big]_{n=0}^{N_i-1}, \tag{4}$$

where $\mathbf{1}_{N_i} \in \mathbb{R}^{N_i\times 1}$ and $\odot$ denotes element-wise multiplication with channel-wise broadcasting (equivalently, $\hat{\mathbf{T}}_i = \mathbf{T}_i\,\mathrm{diag}(\mathbf{a}_i^{\top})$).

C. Cross-Time Reweighting (CT-Reweighting) Module

The token sequences $\widehat{\mathbf{T}}_i$ fused by CS-SE attention are concatenated along the temporal dimension to obtain a global token sequence $\widehat{\mathbf{Z}} \in \mathbb{R}^{T\times H}$. A unified cross-time reweighting operator $\mathrm{CT}(\cdot)$ is then applied to this sequence to obtain the reweighted representation $\mathbf{Z}$, which is fed into the backbone.

$\mathrm{CT}(\cdot)$ is a grouped-channel, dual-branch reweighting module rather than a simple convolution. Specifically, the channels are divided into G groups, and two parallel branches are constructed: (i) a 1×1 convolution that models time–channel contextual gating; and (ii) a 3×3 convolution along the temporal axis that captures local temporal patterns. The outputs of the two branches are fused via an attention-like Softmax–MatMul mechanism to generate temporal weights, which are then used to reweight $\widehat{\mathbf{Z}}$.

To enable global, adaptive control, we introduce a learnable scalar gate, $\gamma = \mathrm{softplus}(\theta)$, to scale the reweighted result. The final input to the backbone and the associated positional information are defined as:

$$\begin{aligned} \widehat{\mathbf{Z}} &= \big[\widehat{\mathbf{T}}_1;\ldots;\widehat{\mathbf{T}}_K\big] \in \mathbb{R}^{T\times H},\\ \mathbf{Z} &= \gamma\,\mathrm{CT}(\widehat{\mathbf{Z}}),\\ \mathrm{pos} &= \big[\mathrm{pos}^{(1)};\ldots;\mathrm{pos}^{(K)}\big],\\ \mathrm{mask} &= \big[\mathrm{mask}^{(1)};\ldots;\mathrm{mask}^{(K)}\big], \end{aligned} \tag{5}$$

where $T = \sum_i N_i$. Subsequently, multi-head self-attention (MHSA) adopts RoPE to encode positional information.

D. Backbone Encoder Layer

The backbone stacks N_L encoder layers. Each layer adopts *Pre-RMSNorm* and sequentially consists of *time-aware self-attention (MHSA+RoPE)* and a *sparse Mixture of Experts (MoE)* block. The layerwise structure is given by

$$\begin{aligned} \mathbf{Z}' &= \mathbf{Z} + \mathrm{SA}_{\mathrm{RoPE}}\big(\mathrm{RMSNorm}(\mathbf{Z});\,\mathrm{pos}\big),\\ \mathbf{H} &= \mathbf{Z}' + \mathrm{MoE}\big(\mathrm{RMSNorm}(\mathbf{Z}')\big). \end{aligned} \tag{6}$$

We use Pre-RMSNorm (ϵ=10^{-6}) instead of LayerNorm as pre-normalization to improve numerical stability. RMSNorm is defined as $\mathrm{RMSNorm}(\mathbf{x};\mathbf{g}) = \mathbf{g}\odot\mathbf{x}/\sqrt{\frac{1}{H}\sum_{i=1}^{H}x_i^2+\epsilon}$, with a learnable per-channel scaling parameter $\mathbf{g}$.

E. MoE with Shared Expert

MoE adopts fixed top-k routing based on Softmax scores and incorporates a shared expert. Routing is performed token-wise on normalized representations:

$$\begin{aligned} \mathbf{p}(\mathbf{x}) &= \mathrm{softmax}\big(\mathrm{RMSNorm}(\mathbf{x})\,\mathbf{W}_g\big),\\ \mathcal{K}(\mathbf{x}) &= \mathrm{Top}\!-\!k\big(\mathbf{p}(\mathbf{x})\big), \quad \mathbf{W}_g \in \mathbb{R}^{H\times E}, \end{aligned} \tag{7}$$

$$\mathrm{MoE}(\mathbf{x}) = \sum_{j\in\mathcal{K}(\mathbf{x})} p_j(\mathbf{x})\,E_j(\mathbf{x}) + \sigma(\mathbf{x}\,\mathbf{w}_s)\,E_s(\mathbf{x}). \tag{8}$$

where $E_s(\cdot)$ denotes the shared expert and $\mathbf{w}_s \in \mathbb{R}^{H\times 1}$ is a learnable gating parameter. The routed experts are selected based on the Top-k Softmax routing scores and model input-dependent specialized degradation patterns, whereas the shared expert captures degradation components common across samples. Therefore, instead of being added with a fixed coefficient, the shared expert is modulated by an independent sigmoid gate, which makes its contribution input-adaptive and bounded. This helps prevent the shared branch from dominating the routed experts while still providing a stable shared basis for prediction. For the k activated experts, the intermediate feed-forward dimension is set to d_{ff}/k (static, with k=`num_experts_per_tok`), while the shared expert keeps the full width.

F. Time-aware Pooling & Prediction

Let the backbone output be $\mathbf{H} = [\mathbf{h}_n]_{n=1}^{T}$, and denote the timestamps by $\mathbf{p} \in \mathbb{R}^{T}$ and the validity masks by $\mathbf{m} \in \{0,1\}^{T}$. We define

$$\begin{aligned} \tilde{p}_n &= \begin{cases} p_n, & m_n = 1,\\ -\infty, & m_n = 0 \end{cases}, \quad p^{\star} = \max_n \tilde{p}_n,\\ \mathcal{S} &= \{n \mid m_n = 1, p_n = p^{\star}\},\\ \mathbf{h} &= \begin{cases} \frac{1}{|\mathcal{S}|}\sum_{n\in\mathcal{S}}\mathbf{h}_n, & |\mathcal{S}| > 0,\\ \mathbf{h}_{\arg\max_n \tilde{p}_n}, & \text{otherwise} \end{cases} \end{aligned} \tag{9}$$

The prediction head is $\hat{y} = \mathbf{h}\mathbf{W} + b$, where $\mathbf{W} \in \mathbb{R}^{H\times 1}$ and $b \in \mathbb{R}$.

G. Loss Function

The main task loss L_{task} is the mean absolute error (MAE), and the total loss is $L = L_{\text{task}} + \lambda_{\mathrm{aux}}L_{\mathrm{aux}}$, where $\lambda_{\mathrm{aux}} > 0$ is the weight of the auxiliary term. To mitigate routing bias, we introduce an MoE load-balancing loss. Let the number of experts at layer ℓ be E, and average the routing probabilities over tokens in the batch to obtain $\bar{\mathbf{p}}^{(\ell)} \in \mathbb{R}^{E}$. Its deviation from the uniform vector $\mathbf{u} = \frac{1}{E}\mathbf{1}$ is defined as

$$L_{\mathrm{aux}}^{(\ell)} = \frac{1}{E}\big\|\bar{\mathbf{p}}^{(\ell)} - \mathbf{u}\big\|_2^2, \qquad L_{\mathrm{aux}} = \frac{1}{N_L}\sum_{\ell=1}^{N_L} L_{\mathrm{aux}}^{(\ell)}. \tag{10}$$

TABLE I
KEY PARAMETERS OF THE NASA, GOTION, AND TJU DATASETS.

Source	Battery	Constant Discharge Current	Rated Capacity	Charge/Discharge Cut-off Voltage
NASA	B0005	2 A	2 Ah	4.2 V / 2.7 V
NASA	B0006	2 A	2 Ah	4.2 V / 2.5 V
NASA	B0007	2 A	2 Ah	4.2 V / 2.2 V
NASA	B0018	2 A	2 Ah	4.2 V / 2.5 V
GOTION	Cell01	27 A	27 Ah	3.65 V / 2.0 V
GOTION	Cell02	27 A	27 Ah	3.65 V / 2.0 V
GOTION	Cell03	27 A	27 Ah	3.65 V / 2.0 V
TJU	CY25-1	2.5 A	2.5 Ah	4.2 V / 2.5 V
TJU	CY25-2	2.5 A	2.5 Ah	4.2 V / 2.5 V
TJU	CY25-3	2.5 A	2.5 Ah	4.2 V / 2.5 V

IV. EXPERIMENTAL SETUP

A. Datasets and Preprocessing

To systematically evaluate the generalization and robustness of the proposed model, we adopt three representative public datasets (key parameters are summarized in Table I, and degradation trajectories are shown in Fig. 2), ranging from small laboratory-scale cells to large commercial cells. We consider two task settings:

(1) *Univariate capacity prediction*, which emulates data-sparse scenarios. This task uses the NASA dataset [29] (LCO cells), a classical benchmark widely used for comparison with existing methods, and the GOTION dataset [30] (27 Ah LFP prismatic cells); the latter is used to evaluate scalability to industrial-grade high-capacity cells.

(2) *Multivariate prediction*, which emulates BMS scenarios with rich measurements and evaluates the model's ability for multi-source feature fusion.

This task employs the TJU dataset [31], which includes both NCM and NCA cells. Following the feature extraction framework proposed by Wang et al. [32], we derive 17 cycle-level health indicators. These include six voltage statistical features (mean, standard deviation, kurtosis, skewness, slope, and entropy), six current statistical features (mean, standard deviation, kurtosis, skewness, slope, and entropy), four charging process metrics (constant-current charge capacity and time, constant-voltage charge capacity and time), and the instantaneous capacity.

Before being fed into the model, all data undergo a unified preprocessing pipeline that includes 3σ-based isolated outlier removal, linear interpolation for the removed points, and normalization. In this work, we only perform abnormal-point cleaning and do not apply a dedicated preprocessing procedure to explicitly identify or correct rest-period-induced effects. Specifically, only isolated abnormal points are removed, while non-isolated local fluctuations are retained in the sequence. Capacity values are normalized as C/C_0, while multivariate features are scaled by Min–Max normalization. All statistics (e.g., min and max) are estimated only on the training set and then reused for validation and test sets to avoid information leakage.

B. Problem Formulation and Evaluation Protocol

Let C_k denote the capacity at the k-th cycle and C_0 the rated capacity. The state of health (SOH) and remaining useful life

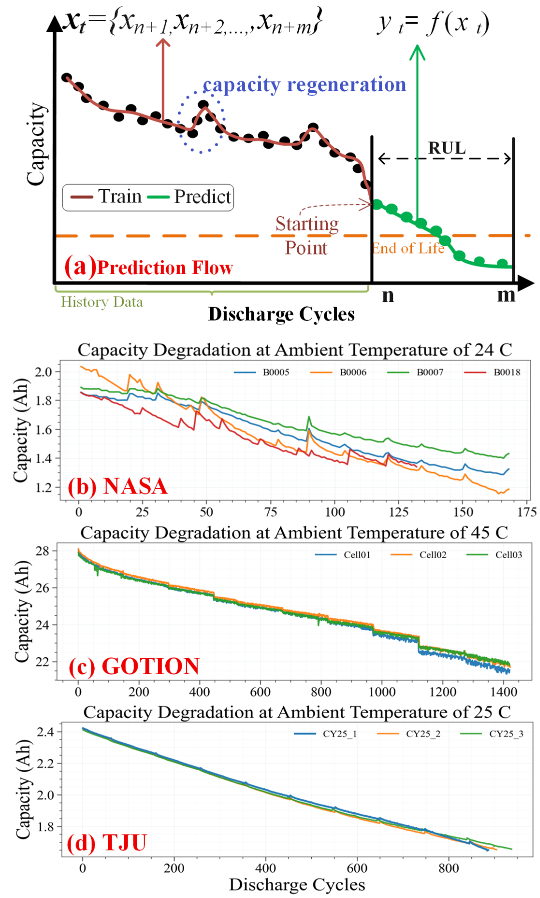

Fig. 2. (a) RUL prediction flowchart; (b) NASA; (c) GOTION; (d) TJU.

(RUL) are defined as

$$\mathrm{SOH}_k = \frac{C_k}{C_0} \times 100\%, \qquad \mathrm{RUL}_k = N_{\mathrm{EOL}} - k. \tag{11}$$

We adopt a sliding-window one-step-ahead prediction framework. At cycle k, the input is a historical window of length L_{in}: for NASA and GOTION, it is the univariate capacity sequence $\{C_{k-L_{\text{in}}+1}, \ldots, C_k\}$; for TJU, it is the multivariate sequence $\{\mathbf{x}_{k-L_{\text{in}}+1}, \ldots, \mathbf{x}_k\}$, where $\mathbf{x}_k \in \mathbb{R}^{17}$ denotes the extracted cycle-level health indicators at cycle k. In all cases, the prediction target is the next-cycle capacity $\hat{C}_{k+1}$. During evaluation, one-step predictions are generated from the corresponding ground-truth historical windows. The predicted RUL is then determined from the sequence of such one-step predictions as the cycle distance to the first point below the threshold τC_0. Accordingly, the reported RE reflects RUL estimation under the adopted one-step evaluation protocol rather than autonomous long-horizon rollout forecasting.

To assess generalization to unseen cells, we use a held-out-cell evaluation protocol. For each dataset, one cell is held out as the test set, and the remaining cells are used for training and validation. The test cell is completely excluded from training, validation, and normalization-statistics estimation. Details of the split are summarized in Table II.

TABLE II
HELD-OUT-CELL DATASET SPLITS (TRAIN/VAL = 80%/20%).

Dataset	Train/Val Cells	Test Cell	EOL Threshold	SP
NASA	B0006, B0007, B0018	B0005	70%	50 / 90
GOTION	Cell02, Cell03	Cell01	80%	450 / 750
TJU	CY25-2, CY25-3	CY25-1	70%	200 / 400

SP denotes early/late prediction start points for RUL evaluation

C. Evaluation Metrics

We evaluate both trajectory fitting and failure-time prediction using MAE, RMSE, R^2, and relative error (RE), defined as

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{12}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{13}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}, \tag{14}$$

$$\mathrm{RE} = \frac{|\mathrm{RUL}_{\mathrm{true}} - \mathrm{RUL}_{\mathrm{pred}}|}{\mathrm{RUL}_{\mathrm{true}}}. \tag{15}$$

D. Baseline Models

To comprehensively benchmark the proposed model, we compare it against a set of representative baselines that cover the main families of deep time-series architectures:

- **ModernTCN** [33]: a TCN-based model designed to capture local and hierarchical temporal patterns in time series.
- **Autoformer** [34]: a Transformer-based forecasting model that models long-range dependencies through an autocorrelation mechanism.
- **iTransformer** [35]: an inverted-dimension Transformer model that treats variables as tokens to model inter-variable dependencies.
- **PatchFormer** [23]: a domain-specific Transformer model designed for battery degradation data.
- **RUL-Mamba** [10]: an SSM-based model developed for battery RUL prediction.

This benchmark suite spans CNN-, Transformer-, and SSM-based architectures and includes both generic time series forecasting models and battery-specific methods, providing a broad comparison.

E. Training and Hyperparameters

The final architectures and training hyperparameters for each dataset are summarized in Table III. Hyperparameters were determined using only the training/validation split of each held-out-cell setting, without using the held-out test cell. We adopted a bounded hyperparameter search over candidate multi-scale patch sets, model-capacity settings, MoE-related parameters, and optimization hyperparameters. We selected the final configuration according to validation performance, training stability, and computational efficiency. All experiments were conducted on a workstation equipped with an NVIDIA RTX 3060 GPU and an Intel Core i5-13490F CPU using PyTorch 2.1.0.

TABLE III
MODEL ARCHITECTURE AND TRAINING HYPERPARAMETERS.

Hyperparameter	NASA	GOTION	TJU
Architecture			
Patch lengths (cycles) $\mathcal{P}$	{2,4,8}	{8,16,32}	{4,8,16}
Look-back length L_{in}	16	64	64
Embedding dim d_{model}	64	128	256
Encoder layers L	1	1	2
FFN hidden dim d_{ff}	128	512	1024
MoE experts E	4	8	4
Top-k (gated) k	2	2	2
Router balance weight λ_{aux}	0.001	0.001	0.01
Dropout rate p_{drop}	0.05	0.10	0.10
Optimization / Training			
Optimizer	Adam	Adam	Adam
Learning rate η	1e^{-3}	1e^{-3}	1e^{-3}
Batch size B	128	128	128
Early stopping (metric / patience)	MAE / 10	MAE / 10	MAE / 10
Params	**94.5K**	**1.1M**	**5.4M**

Notes. The table reports the final selected configurations for each dataset. During hyperparameter tuning, candidate multi-scale patch sets were explored for $\mathcal{P}$ (e.g., $\{2,4,8\}$ and $\{1,2,4,8\}$), while the search ranges for other tunable parameters were $d_{\text{model}} \in [16, 256]$, $L \in [1, 4]$, $d_{\text{ff}} \in [16, 1024]$, $E \in [4, 16]$, $k \in [1, 3]$, $\lambda_{\text{aux}} \in [0, 0.1]$, $p_{\text{drop}} \in [0, 0.2]$, $\eta \in [10^{-4}, 10^{-2}]$, and $B \in [64, 512]$.

V. RESULTS AND DISCUSSION

All results in this section are averaged over ten independent runs with different random seeds to ensure statistical reliability.

A. Comparative Performance Analysis

Table IV summarizes the predictive performance of BATTER-MoE and the competing baselines on the NASA, GOTION, and TJU datasets under both early and late start-point settings. Across these heterogeneous benchmarks, BATTER-MoE achieves the best or tied-best performance in most settings, with the lowest MAE, RMSE, and RE in nearly all cases and the highest or tied-highest R^2.

The differences between methods become more apparent in the resulting RUL estimation under the adopted evaluation protocol. On the NASA dataset, the relative error is reduced to 0 under both early and late settings. On the GOTION dataset, BATTER-MoE also substantially reduces RE compared with the strongest competing methods, for example, by 79.15% under the Early setting, indicating more accurate end-of-life localization on long-life, slowly degrading cells. Similar improvements are observed on the multivariate TJU dataset, where BATTER-MoE achieves the best MAE and RMSE under both start points. Taken together, these results suggest that the proposed framework remains effective under both heterogeneous data conditions and different prediction start points.

The observed gains are consistent with the BATTER-MoE design. Compared with dense baselines that process all inputs through a shared computation pathway, the proposed framework combines battery-oriented multi-scale temporal representation learning with sparse conditional computation. This design is well-suited to battery degradation data, which often

TABLE IV
RESULTS ON NASA, GOTION, AND TJU WITH EARLY (50/450/200) AND LATE (90/750/400) START POINTS.

Method	Start	NASA(Univariate)				GOTION(Univariate)				TJU(Multivariate)			
		MAE	RMSE	R^2	RE	MAE	RMSE	R^2	RE	MAE	RMSE	R^2	RE
ModernTCN	Early	0.0069	0.0101	0.9940	0.0114	0.0553	0.0750	0.9960	0.0365	0.0022	0.0029	0.9996	0.0040
Autoformer	Early	0.0234	0.0336	0.9341	0.0486	0.0694	0.0980	0.9930	0.0548	0.0020	0.0028	0.9997	0.0028
iTransformer	Early	0.0085	0.0116	0.9922	0.0257	0.0564	0.0770	0.9957	0.0379	0.0018	0.0025	0.9997	0.0064
PatchFormer	Early	0.0073	0.0104	0.9937	0.0171	0.0544	0.0737	0.9961	0.0283	0.0015	0.0022	0.9998	0.0035
RUL-Mamba	Early	0.0083	0.0134	0.9901	0.0135	0.0602	0.0841	0.9949	0.0361	0.0015	0.0025	0.9997	0.0054
Ours	Early	**0.0045**	**0.0088**	**0.9956**	**0.0000**	**0.0473**	**0.0640**	**0.9970**	**0.0059**	**0.0011**	**0.0017**	**0.9999**	**0.0022**
Gain vs 2nd-best (%)	Early	**34.78%**	**12.87%**	**0.16%**	**100.00%**	**13.05%**	**13.16%**	**0.09%**	**79.15%**	**26.67%**	**22.73%**	**0.01%**	**21.43%**
ModernTCN	Late	0.0066	0.0096	0.9806	0.0200	0.0619	0.0829	0.9921	0.1304	0.0025	0.0032	0.9991	0.0061
Autoformer	Late	0.0213	0.0313	0.8153	0.1235	0.0783	0.1082	0.9864	0.1739	0.0038	0.0051	0.9975	0.0116
iTransformer	Late	0.0077	0.0106	0.9764	0.1133	0.0642	0.0868	0.9913	0.1319	0.0025	0.0032	0.9991	0.0095
PatchFormer	Late	0.0065	0.0097	0.9804	0.0533	0.0630	0.0854	0.9916	0.1232	0.0019	0.0026	0.9994	0.0079
RUL-Mamba	Late	0.0092	0.0161	0.9556	0.0735	0.0683	0.0943	0.9898	0.1145	0.0017	0.0026	0.9994	0.0078
Ours	Late	**0.0046**	**0.0084**	**0.9854**	**0.0000**	**0.0535**	**0.0716**	**0.9941**	**0.0188**	**0.0013**	**0.0019**	**0.9998**	**0.0037**
Gain vs 2nd-best (%)	Late	**29.23%**	**12.50%**	**0.49%**	**100.00%**	**13.57%**	**13.63%**	**0.20%**	**83.58%**	**23.53%**	**26.92%**	**0.04%**	**39.34%**

contain coexisting long-term fade trends, local regeneration-like fluctuations, and cross-dataset variability. Relative to convolution-based models such as ModernTCN, BATTER-MoE provides a mechanism to model both global degradation behavior and local temporal irregularities. Compared with dense Transformer-based and state-space-based models, the proposed framework introduces time-aware multi-scale tokenization, cross-scale feature fusion, and temporal reweighting before backbone encoding, thereby yielding representations that encode degradation dynamics more explicitly across temporal scales. In addition, sparse routing provides a mechanism for adapting computation to different input patterns. At the same time, time-aware pooling helps focus prediction on the latest valid timestamp, which is particularly relevant for RUL estimation across different start-point settings.

Fig. 4 provides a visual comparison of prediction errors and degradation trajectories. The predicted curves produced by BATTER-MoE are closer to the ground-truth trajectories across different datasets and start points. At the same time, the associated error bars are generally smaller than those of the baselines. This observation is consistent with the quantitative results in Table IV and indicates that BATTER-MoE achieves lower average error and more stable predictions across repeated runs. As a complementary exploratory analysis on the multivariate TJU dataset, Fig. 3 reports normalized gradient-based feature importance scores. For each run and start-point setting, feature importance is estimated on the evaluation samples, aggregated across batches, and then normalized within that run for visualization. The results show that BATTER-MoE consistently assigns relatively greater importance to a limited subset of input features across different runs and start points, indicating concentrated and repeatable feature usage.

For additional reference on the classical NASA benchmark, Table V compares BATTER-MoE with representative methods previously reported for battery B0005. Among the methods reported at the same start point of 50 cycles, BATTER-MoE achieves the lowest MAE and RMSE while maintaining zero RE. It also outperforms the prior MoE-based method AttMoE in this setting, providing additional evidence that the framework remains competitive on a widely used benchmark.

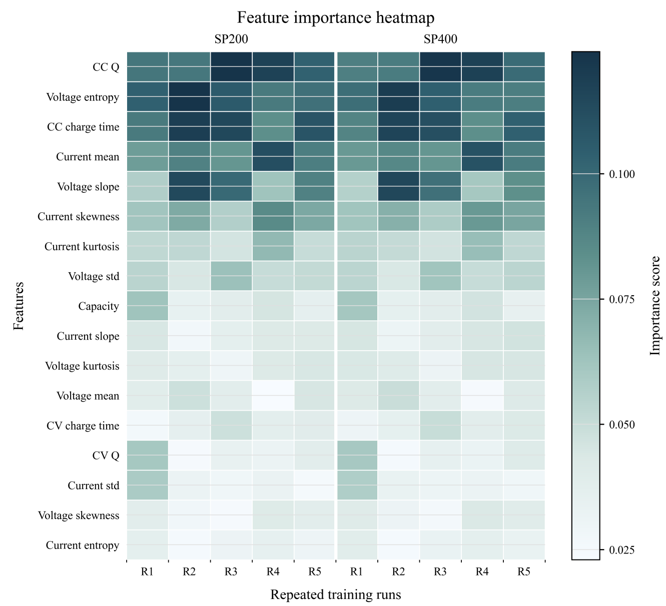

Fig. 3. Normalized gradient-based feature importance on the TJU dataset under SP200 and SP400 over five repeated runs (R1–R5).

TABLE V
COMPARISON ON THE NASA B0005 TEST CELL.

Battery	Method	Year	SP	MAE	RMSE	RE
B0005	PSO multi-kernel RVM [36]	2022	40	0.0258	0.0351	0.0290
	TCN-LSTM [37]	2023	50	0.0152	0.0203	0.0133
	TCN-GRU-DNN [38]	2023	60	0.0063	0.0125	0.0000
	AttMoE [28]	2024	50	0.0096	0.0164	0.0112
	RUL-Mamba [10]	2025	50	0.0083	0.0134	0.0135
	Ours	-	50	0.0045	0.0088	0.0000

B. Synergistic Effects of Core Components

To quantify the contribution of each component, we conduct ablation studies on NASA (SP = 50), GOTION (SP = 450), and TJU (SP = 200), as summarized in Table VI.

The results show that multi-scale tokenization and the proposed cross-scale/cross-time modules contribute in a complementary manner. The *w/o Patch* variant consistently degrades performance on all three datasets, indicating that patch tokenization provides a stronger basis for subsequent routing

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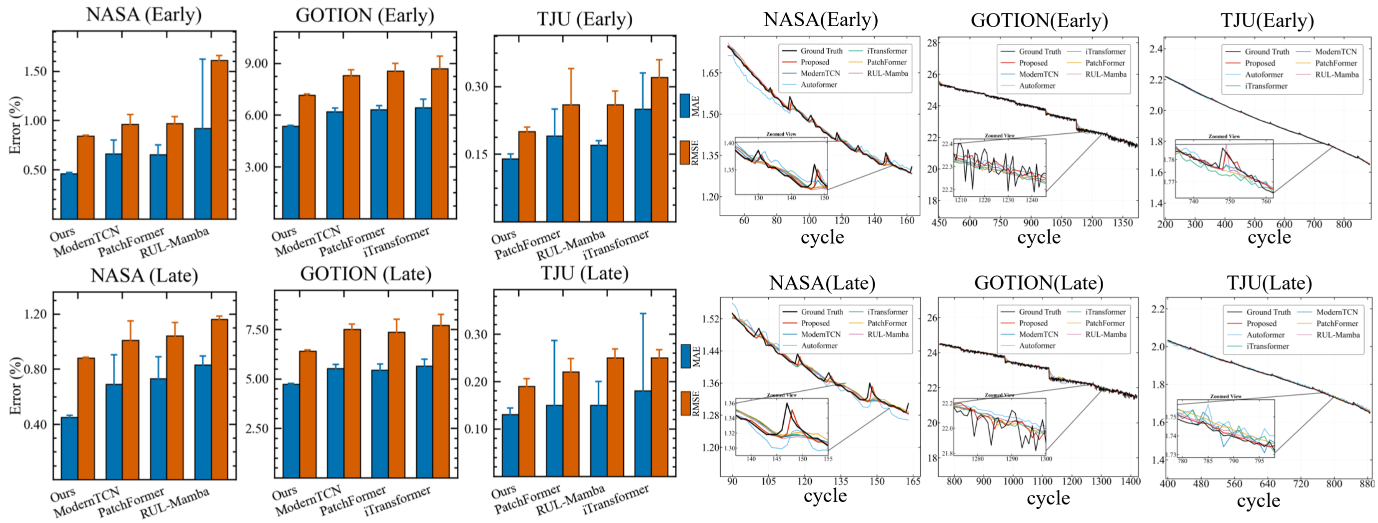

Fig. 4. Prediction results of BATTER-MoE on multiple datasets. Left: MAE/RMSE comparison; right: capacity degradation curves. Error bars denote the standard deviation over 10 independent runs with different random seeds.

TABLE VI
ABLATION STUDY OF BATTER-MOE. PARENTHESES SHOW RELATIVE REDUCTION (%) OF THE FULL MODEL VS. EACH ABLATED VARIANT.

	NASA (Univariate)			GOTION (Univariate)			TJU (Multivariate)		
Method	**MAE**	**RMSE**	**RE**	**MAE**	**RMSE**	**RE**	**MAE**	**RMSE**	**RE**
w/o CS-SE	0.0074 (39)	0.0129 (32)	0.0070 (100)	0.0610 (22)	0.0715 (10)	0.0245 (76)	0.0019 (42)	0.0024 (29)	0.0051 (57)
Classic SE	0.0063 (29)	0.0098 (10)	0.0000 (—)	0.0616 (23)	0.0721 (11)	0.0245 (76)	0.0017 (35)	0.0024 (29)	0.0038 (42)
w/o CT-Reweighting	0.0068 (34)	0.0109 (19)	0.0090 (100)	0.0630 (25)	0.0744 (14)	0.0257 (77)	0.0016 (31)	0.0023 (26)	0.0044 (50)
w/o CS-SE and CT-Reweighting	0.0089 (49)	0.0138 (36)	0.0133 (100)	0.0712 (34)	0.0819 (22)	0.0268 (78)	0.0022 (50)	0.0028 (39)	0.0062 (65)
w/o MoE	0.0062 (27)	0.0098 (10)	0.0000 (—)	0.0636 (26)	0.0728 (12)	0.0180 (67)	0.0018 (39)	0.0025 (32)	0.0041 (46)
w/o Patch	0.0055 (18)	0.0093 (5)	0.0000 (—)	0.0504 (6)	0.0687 (7)	0.0110 (46)	0.0017 (35)	0.0023 (26)	0.0047 (53)
Mean Pooling	0.0049 (8)	0.0092 (4)	0.0000 (—)	0.0510 (7)	0.0697 (8)	0.0104 (43)	0.0015 (27)	0.0021 (19)	0.0038 (42)
Index Pooling	0.0050 (10)	0.0093 (5)	0.0000 (—)	0.0521 (9)	0.0667 (4)	0.0145 (59)	0.0014 (21)	0.0021 (19)	0.0050 (56)
Sequential-index RoPE	0.0049 (8)	0.0089 (1)	0.0000 (—)	0.0483 (2)	0.0648 (1)	0.0080 (26)	0.0014 (21)	0.0021 (19)	0.0035 (37)
w/o RoPE	0.0051 (12)	0.0090 (2)	0.0028 (100)	0.0536 (12)	0.0703 (9)	0.0145 (59)	0.0015 (27)	0.0022 (23)	0.0052 (58)
Last-token Pooling	0.0052 (13)	0.0094 (6)	0.0000 (—)	0.0566 (16)	0.0691 (7)	0.0069 (14)	0.0014 (21)	0.0020 (15)	0.0035 (37)
Full Model	**0.0045**	**0.0088**	**0.0000**	**0.0473**	**0.0640**	**0.0059**	**0.0011**	**0.0017**	**0.0022**

We omit R^2 in ablations due to saturation near 0.99 and low sensitivity. NASA (SP=50), GOTION (SP=450), and TJU (SP=200).

and prediction. Both *w/o CS-SE* and *w/o CT-Reweighting* lead to noticeable performance degradation, while *w/o CS-SE + CT-Reweighting* causes the largest drop. This suggests that CS-SE and CT-Reweighting play distinct but complementary roles: the former performs cross-scale channel recalibration, whereas the latter performs adaptive temporal reweighting. Together, they improve the modeling of multi-scale degradation dynamics.

The comparison with *Classic SE* further confirms the importance of explicit cross-scale interaction. As shown in Fig. 5, classic SE performs channel reweighting within a single scale, whereas CS-SE first aggregates information across scales and then recalibrates channels. Consistently, the *Classic SE* variant outperforms *w/o CS-SE*, but remains inferior to the full model. This indicates that channel reweighting alone is beneficial, while explicit cross-scale aggregation provides additional gains by better balancing long-term degradation trends and short-term fluctuations.

We further evaluate time encoding using two RoPE variants. The *Sequential-index RoPE* variant retains RoPE but replaces temporal positions with uniformly spaced sequential indices, whereas *w/o RoPE* removes rotary positional encoding entirely. The full model consistently achieves the best results, indicating that RoPE is effective and that temporal-position information provides additional benefits beyond token order alone. This improvement is more pronounced on GOTION and TJU, suggesting that time-aware positional encoding is particularly useful for longer and more heterogeneous degradation trajectories.

The MoE backbone and time-aware pooling are also important for prediction accuracy. The *w/o MoE* variant increases prediction error on all datasets, confirming that sparse expert routing improves adaptation to heterogeneous degradation patterns. At the readout stage, *Mean Pooling*, *Index Pooling*, and *Last-token Pooling* all underperform the proposed time-aware pooling, showing that focusing on the latest valid timestamp provides a more reliable summary for RUL prediction under the adopted multi-scale tokenization setting.

Overall, the observed gains of BATTER-MoE are associated with the joint effects of multi-scale tokenization, CS-SE, CT-Reweighting, sparse MoE routing, RoPE-based time encoding, and time-aware pooling, rather than with any individual component in isolation.

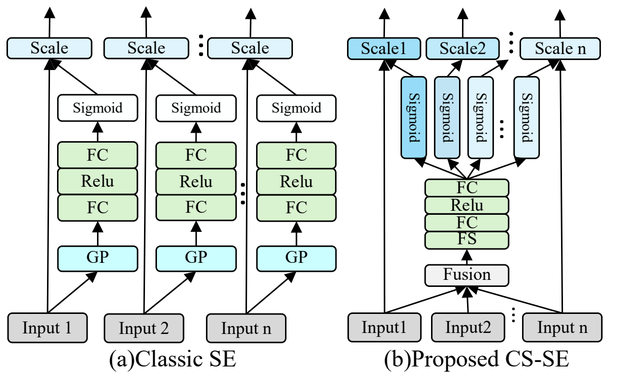

Fig. 5. Comparison between the *Classic SE* block and the proposed *CS-SE* block.

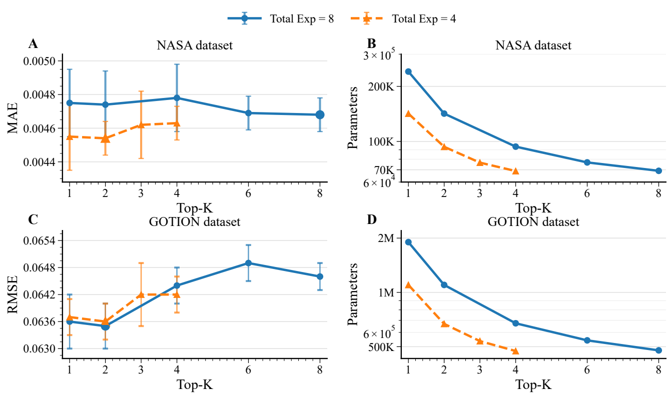

Fig. 6. Impact of Top-k routing on prediction error (MAE and RMSE) and total parameter count for the NASA and GOTION datasets.

C. Optimization of MoE Key Parameters

We further investigate the effect of Top-k, a key hyperparameter in the MoE module, on predictive performance and parameter efficiency. The results are shown in Fig. 6, where the blue solid and orange dashed curves denote configurations with 8 and 4 total experts, respectively.

In terms of predictive performance, the model achieves the lowest error and the smallest variance when $k = 1$ or 2 on both datasets. Specifically, the best MAE on NASA is observed in Fig. 6A, while the best RMSE on GOTION is shown in Fig. 6C. When $k > 2$, the error increases noticeably, indicating degraded predictive accuracy and stability.

In terms of parameter efficiency, the total number of parameters decreases as k increases on both datasets, as shown in Fig. 6B and D. This trend is consistent with our MoE design: for the activated experts, the intermediate feed-forward dimension is statically set to d_{ff}/k, so a larger k leads to narrower expert sub-networks and hence fewer parameters. Considering both accuracy and efficiency, $k = 2$ provides the most favorable trade-off across the two datasets.

For example, on the GOTION dataset with 8 total experts, the RMSE remains relatively stable, whereas the parameter count drops from approximately 1.9M to 1.1M, corresponding to a reduction of about 40%. Therefore, $k = 2$ is adopted as the default setting in the main experiments reported in Table IV.

D. Parameter Efficiency Under Resource-Constrained Settings

In addition to predictive accuracy, practical battery management applications also require models with a favorable computational profile under limited memory and compute budgets. Table VII reports the training and inference time of different methods on the three datasets. Under the current experimental settings, BATTER-MoE attains the lowest training time on all three benchmarks, requiring 4.69 s on NASA, 16.54 s on GOTION, and 43.67 s on TJU. Its inference latency is also the lowest or among the lowest across the compared methods, while the average stopping epoch remains relatively small. Taken together, these results suggest that BATTER-MoE can achieve competitive predictive performance with comparatively efficient optimization and inference under the same early-stopping protocol.

The timing differences in Table VII should be interpreted with caution, as all compared models are relatively lightweight and the benchmarks are of moderate scale. Therefore, BATTER-MoE is better understood as achieving a favorable balance between predictive performance and computational efficiency.

To further characterize the deployment cost of BATTER-MoE, Table VIII reports the main computational indicators under the primary experimental configurations. The model maintains a compact profile across all three datasets, with FLOPs ranging from 3.81×10^5 to 1.35×10^7 and peak runtime memory usage from 9.44 MB to 20.37 MB. The measured batch-1 inference latency remains below 9 ms on both GPU and CPU platforms, even for the most complex TJU setting. The relatively similar CPU and GPU latency can be partly attributed to the compact model size and the batch-1 setting, in which GPU kernel-launch overhead can dominate execution time. These results indicate that BATTER-MoE maintains competitive predictive accuracy with a practical runtime footprint, which is relevant for deployment-oriented battery prognostics scenarios.

We further examine scalability under stricter parameter budgets by constructing reduced-size variants of BATTER-MoE, as shown in Table IX. The results show that the model remains competitive even after substantial parameter compression. On NASA, the 1.34K-parameter variant still achieves an MAE of 0.0056 and an RMSE of 0.0094, which remain close to the full model. On GOTION, the 5.77K-parameter variant achieves an MAE of 0.0502 and an RMSE of 0.0683, again showing only moderate degradation despite the much smaller model size. As the parameter budget increases, performance improves gradually, suggesting a smooth accuracy-parameter trade-off rather than an abrupt dependence on large-scale configurations.

Overall, the results in Tables VII–IX indicate that BATTER-MoE maintains an efficient computational profile in multiple respects: under the current benchmark settings, it trains and runs efficiently, maintains a compact deployment profile, and retains competitive predictive performance under highly compressed model configurations. These properties suggest that the proposed framework can serve as a practical option for battery prognostics scenarios in which both predictive reliability and computational efficiency are important considerations.

TABLE VII
TRAINING AND INFERENCE TIME PER METHOD ON DIFFERENT DATASETS.

Dataset	Method	Epoch	Training time (s)	Inference time (s)
NASA	ModernTCN	52	9.52	0.10
	Autoformer	32	8.00	0.10
	iTransformer	63	5.06	0.09
	PatchFormer	31	10.95	0.10
	RUL-Mamba	24	8.30	0.12
	Ours	**22**	**4.69**	**0.09**
GOTION	ModernTCN	32	21.62	0.56
	Autoformer	89	126.58	0.65
	iTransformer	46	41.88	0.59
	PatchFormer	28	51.29	0.59
	RUL-Mamba	49	187.62	0.58
	Ours	**23**	**16.54**	**0.55**
TJU	ModernTCN	52	65.77	0.64
	Autoformer	182	214.45	0.73
	iTransformer	49	61.66	0.60
	PatchFormer	63	82.95	0.69
	RUL-Mamba	39	91.34	0.66
	Ours	**42**	**43.67**	**0.46**

Note: The SP values are 50 for NASA, 450 for GOTION, and 200 for TJU. Epoch denotes the average stopping epoch over 10 independent runs under the early-stopping strategy.

TABLE VIII
COMPUTATIONAL CHARACTERISTICS OF BATTER-MOE UNDER THE MAIN EXPERIMENTAL CONFIGURATIONS.

Dataset	Params	FLOPs	GPU Latency (ms)	CPU Latency (ms)	Peak Memory (MB)
NASA	94.5K	3.81×10^5	4.04	3.80	9.44
GOTION	1.1M	1.94×10^6	5.38	5.14	10.77
TJU	5.4M	1.35×10^7	8.83	8.56	20.37

Note: Latency is measured with batch size = 1.

E. Robustness to Noisy Training Inputs

To evaluate robustness to imperfect training data, we injected synthetic mixed noise (30% burst noise and 70% Gaussian noise) into the NASA training set at corruption ratios of 0, 0.1, and 0.2, as reported in Table X. Traditional machine learning baselines such as SVR and GPR achieved reasonable accuracy on clean data. However, their prediction errors increased markedly as the noise level rose, indicating limited robustness to corrupted observations. Deep learning baselines, including PatchFormer and RUL-Mamba, exhibited better stability, yet remained consistently inferior to BATTER-MoE. Even at the highest corruption ratio of 0.2, BATTER-MoE still achieved the lowest MAE (0.0077) and RMSE (0.0112) among all compared methods. These results indicate that BATTER-MoE exhibits less performance degradation than the compared methods under the tested noise settings, which is relevant to battery prognostics scenarios with imperfect measurements.

F. Transfer Learning with Limited Target Data

To evaluate the effect of transfer learning with limited target-domain data, we conducted additional experiments on the GOTION dataset with training data ratios ranging from 0.1 to 0.5. BATTER-MoE was first pretrained on NASA and then fully fine-tuned on GOTION without freezing any parameters. For fairness, the scratch model used the same GOTION hyperparameter setting as in Table III.

TABLE IX
PERFORMANCE OF BATTER-MOE UNDER VARYING PARAMETER BUDGETS.

Dataset	Params	L	d_{model}	d_{ff}	MAE	RMSE
NASA	1.34K	1	8	16	0.0056	0.0094
	10.16K	2	8	64	0.0050	0.0090
	94.5K	1	64	128	0.0045	0.0088
GOTION	5.77K	1	16	16	0.0502	0.0683
	36.78K	1	32	64	0.0484	0.0643
	1.1M	1	128	512	0.0473	0.0640

Note: L denotes the number of encoder layers, d_{model} the hidden dimension, and d_{ff} the feed-forward dimension.

TABLE X
ROBUSTNESS UNDER NOISY TRAINING INPUTS ON THE NASA DATASET.

Method	Noise Ratio	MAE	RMSE	RE
SVR	0.0	0.0103	0.0128	0.0278
SVR	0.1	0.0235	0.0312	0.0694
SVR	0.2	0.0291	0.0344	0.0733
GPR	0.0	0.0095	0.0122	0.0139
GPR	0.1	0.0164	0.0209	0.0185
GPR	0.2	0.0261	0.0347	0.0602
PatchFormer	0.0	0.0073	0.0104	0.0171
PatchFormer	0.1	0.0087	0.0117	0.0157
PatchFormer	0.2	0.0090	0.0118	0.0200
RUL-Mamba	0.0	0.0083	0.0134	0.0135
RUL-Mamba	0.1	0.0088	0.0116	0.0160
RUL-Mamba	0.2	0.0094	0.0118	0.0143
Ours	0.0	**0.0045**	**0.0088**	**0.0000**
Ours	0.1	**0.0066**	**0.0099**	**0.0064**
Ours	0.2	**0.0077**	**0.0112**	**0.0078**

As shown in Table XI, transfer learning consistently improved performance across all training-data ratios, with the largest gains in the low-data regime. For example, at a train ratio of 0.1, the mean MAE decreased from 0.0765 to 0.0532, and the mean RMSE decreased from 0.0892 to 0.0717. In addition, the scratch model showed noticeably larger variance, whereas transfer learning remained much more stable. Although the performance gap gradually narrowed as more target-domain data became available, transfer learning remained consistently superior. These results suggest that pretraining on NASA may improve adaptation efficiency and predictive stability on GOTION when target-domain data are limited.

TABLE XI
TRANSFER LEARNING RESULTS ON THE GOTION DATASET UNDER DIFFERENT TARGET-DOMAIN TRAINING-DATA RATIOS.

Train Ratio	MAE		RMSE	
	Scratch	Transfer	Scratch	Transfer
0.1	0.0765 ± 0.0066	$\mathbf{0.0532 \pm 0.0001}$	0.0892 ± 0.0079	$\mathbf{0.0717 \pm 0.0001}$
0.2	0.0668 ± 0.0026	$\mathbf{0.0523 \pm 0.0006}$	0.0798 ± 0.0037	$\mathbf{0.0695 \pm 0.0007}$
0.3	0.0617 ± 0.0014	$\mathbf{0.0486 \pm 0.0015}$	0.0724 ± 0.0026	$\mathbf{0.0658 \pm 0.0020}$
0.5	0.0542 ± 0.0002	$\mathbf{0.0475 \pm 0.0001}$	0.0754 ± 0.0050	$\mathbf{0.0645 \pm 0.0001}$

Note: "Scratch" denotes training directly on GOTION without pretrained initialization, whereas "Transfer" denotes full fine-tuning from NASA pretrained weights without freezing any parameters. SP is fixed at 450 for all settings.

VI. CONCLUSION

In this work, we proposed BATTER-MoE, a sparse battery-oriented framework for lithium-ion battery RUL prediction that combines multi-scale temporal representation learning with conditional expert routing. Experiments on the NASA, GOTION, and TJU datasets show that BATTER-MoE achieves competitive predictive performance across heterogeneous settings. At the same time, ablation studies confirm the contributions of the proposed temporal modules and sparse routing strategy. These results suggest that sparse conditional computation is a practical direction for battery prognostics when both predictive accuracy and computational efficiency are important. Future work will further examine long-horizon forecasting, broader real-world operating conditions, and the integration of physics-based priors for deployment-oriented battery management.

Acknowledgment

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